

SEAN

✉ zxzhou0916@163.com · ☎ (+86) 185-1985-1160 · 🔗 [homepage](#) · in [zxzhou](#) · G [scholar](#)

🎓 EDUCATION

Rutgers, the State University of New Jersey, USA, *Ph.D.* 2012.09 – 2018.01
Tongji University, Shanghai, *B.S.* 2008.09 – 2012.06

👤 WORK EXPERIENCE

Multimodal Model Division ([HunYuan](#)), Tencent 2024.05 – Present
Senior Researcher – Key Tech Lead of Video Generation & Digital Human

Multi-Subject Consistent Video Generation (Project S)

- Led technical design of multi-subject consistent video generation based on MTSS + VLM + Wan-2.2. Proposed VLM-to-T5 alignment scheme for fine-grained multimodal understanding, and RoPEShift + MemTok injection for high-consistency multi-subject (person/object/scene) generation.
- Built data pipeline handling tens of millions of samples for reference image generation, image-text matching, and augmentation with hard-copy artifact mitigation.

Multimodal Video-Driven Generation

- **Multimodal VideoChat:** Directed a unified framework for joint audio-video generation, audio-driven generation, and video dubbing — the first unification of all three tasks with SOTA performance. 🔗 [UniAVGen](#) - CVPR 2026
- **Human-Object Interaction Digital Human:** Led the first fully multimodal-driven HOI digital human supporting voice, motion, and product trajectory signals. Built HOI data pipeline via VLM + SAM + DINO + DepthAnything. 🔗 [HOMA](#) - SIGGRAPH ASIA 2025
- **Audio-Driven Digital Human:** Developed HunyuanVideo-based system covering real/anime, head-shoulder/full-body modes. Deployed on Tencent Music, Yuanbao, and WeChat Channels. Curated 1,500+ hours of high-quality data. 🔗 [HunyuanVideo-Avatar](#)
- **Video Post-Training:** Led reward model optimization across deformity, physical plausibility, and aesthetics. Proposed Cross-Prompt Pairing, BT-T training, and HPQA strategies. 🔗 [SoliReward](#) - CVPR 2026

2D Digital Human Animation

- **DiT-based:** Led multimodal 2D digital human animation achieving audio-driven, pose-driven, and expression+pose-driven modes. Launched on Yuanbao APP; pose-driven featured on [2025 CCTV Online Spring Festival Gala](#); expression+pose deployed for QQ sticker packs (industry first). Built fully automated data cleaning and multimodal labeling pipeline. 🔗 [HunyuanVideo](#)
- **SVD-based:** Led SVD-based digital human with audio/pose/expression/full-condition modes, deployed across advertising, cloud services, and sticker packs. 🔗 [HunyuanPortrait](#) - CVPR 2025 | 🔗 [ACTalker](#) - ICCV 2025

[XiaoIce](#) xNext Vision Technology / Foundation Model Group 2021.09 – 2024.05
Researcher

Multimodal-Driven Virtual Human Motion

- **AvatarGPT:** LLM-based motion generation & task planning agent supporting generation, understanding, in-betweening, planning, and scene understanding. Proposed V-LLM + ChatGPT automated annotation for training data. 🔗 [AvatarGPT](#) - CVPR 2024
- **UDE-2:** First multimodal (text/music/speech) full-body motion generation model with hierarchical VQ-VAE, parameter re-initialization, and semantic-enhanced loss. 🔗 [UDE-2](#)

- **UDE:** First multimodal-driven motion generation model with pretrained-model-based feature fusion and diffusion-based motion decoder, achieving SOTA.  [UDE](#) - CVPR 2023
- **Evaluation Metrics:** Contrastive learning-based metrics for semantic relevance, style-rhythm consistency (music), and identity consistency (speech).  [MDSC](#)
- **Motion Dataset:** Built video-based full-body pose estimation + global trajectory optimization + occlusion in-betweening pipeline.  [GOMotion](#)

Huawei Autonomous Driving Solution (ADS)

2020.10 – 2021.09

Senior Engineer – AVP Perception Team Lead

- Multi-task perception on 4-camera fisheye stitched images: 18-class segmentation at 10ms on MDC; heatmap + offset regression keypoint detection with $\leq 5\text{cm}$ accuracy.
- Monocular semantic segmentation + depth estimation with scale consistency loss, achieving centimeter-level reconstruction in close-range scenes.
- Lightweight YOLACT-based feature detection and instance segmentation for underground parking global localization.

Baidu Intelligent Driving Group (IDG)

2018.03 – 2020.10

Senior Engineer – HD Map Team Lead

- Designed attention-based point cloud segmentation models with modular + parallelizable deployment for ground feature recognition.
- High-precision feature vectorization and road topology modeling (lane lines, drivable boundaries) from point cloud segmentation.
- GCN-based HD map feature matching: encoding local/global topology for robust multi-pass matching with initial pose offsets $> 10\text{m}$.

Bentley Systems, CT, USA

2017.06 – 2017.09

R&D Engineer (Intern)

- Point cloud-based complex pipe segmentation and modeling for industrial plants. Oral at 2017 Bentley Research Seminar.

Center for Advanced Infrastructure and Transportation (CAIT), NJ, USA

2014.01 – 2018.01

Research Assistant

- USDOT: Vehicle-mounted LiDAR platform and point cloud algorithms for mapping, pipeline detection, and disaster assessment.
- NSF: Multi-sensor fusion indoor navigation system, tested at NYC Blind Hospital and Port Authority transit hubs.

SELECTED PUBLICATIONS

-
- Zhang, Y., **Zhou, Z.**, et al. “Making Avatars Interact: Towards Text-Driven Human-Object Interaction for Controllable Talking Avatars” arXiv:2602.01538, 2026.
 - Sun, Z., **Zhou, Z.**, et al. “StreamAvatar: Streaming Diffusion Models for Real-Time Interactive Human Avatars” arXiv:2512.22065, 2025.
 - Mi, X., **Zhou, Z.**, et al. “Video Generation Models Are Good Latent Reward Models.” arXiv:2511.21541, 2025.
 - Lian, J., **Zhou, Z.**, et al. “SoliReward: Mitigating Susceptibility to Reward Hacking and Annotation Noise in Video Generation Reward Models.” arXiv:2512.22170, 2025.
 - Wu, B., **Zhou, Z.**, et al. “HunyuanVideo 1.5 Technical Report.” arXiv:2511.18870, 2025.
 - Zhang, G., **Zhou, Z.**, et al. “UniAVGen: Unified Audio and Video Generation with Asymmetric Cross-Modal Interactions.” arXiv:2511.03334, 2025.

- Huang, Z., **Zhou, Z.**, et al. “HunyuanVideo-HOMA.” arXiv:2506.08797, 2025.
- Chen, Y., **Zhou, Z.**, et al. “HunyuanVideo-Avatar.” arXiv:2505.20156, 2025.
- Hong, F.-T., **Zhou, Z.**, et al. “Audio-visual controlled video diffusion with masked selective state spaces modeling.” arXiv:2504.02542, 2025.
- Xu, Z., **Zhou, Z.**, et al. “Hunyuanportrait: Implicit condition control for enhanced portrait animation.” CVPR, 2025.
- Kong, W., **Zhou, Z.**, et al. “Hunyuanvideo: A systematic framework for large video generative models.” arXiv:2412.03603, 2024.
- **Zhou, Z.**, Wan, Y., Wang, B. “AvatarGPT: All-in-One Framework for Motion Understanding, Planning, Generation and Beyond.” **CVPR 2024**.
- **Zhou, Z.**, Wan, Y., Wang, B. “A Unified Framework for Multimodal, Multi-Part Human Motion Synthesis.” arXiv:2311.16471.
- **Zhou, Z.**, Wang, B. “MDSC: Towards Evaluating the Style Consistency Between Music and Dance.” arXiv:2309.01340, 2023.
- **Zhou, Z.**, Wang, B. “UDE: A Unified Driving Engine for Human Motion Generation.” **CVPR 2023**.
- **Zhou, Z.**, Gong, J., Hu, X. “Community-scale multi-level post-hurricane damage assessment.” Automation in Construction 98, 2019.
- **Zhou, Z.**, Gong, J. “Automated analysis of mobile LiDAR data for component-level damage assessment.” CACAIE 33.5, 2018.
- **Zhou, Z.**, Gong, J. “Automated residential building detection from airborne LiDAR data with DNNs.” AEI 36, 2018.
- **Zhou, Z.**, Gong, J., Guo, M. “Image-based 3D reconstruction for posthurricane building damage assessment.” JCE 30.2, 2016.
- **Zhou, Z.**, et al. “Multiresolution change analysis for postdisaster natural gas pipeline risk.” TRR 2595.1, 2016.

⚙️ SKILLS

- **C++:** PCL, OpenCV, Eigen, Ceres | **Python:** PyTorch, TensorFlow, PyTorch3D, scikit-learn | **Other:** $\LaTeX$
- **Languages:** English (Fluent), Chinese (Native)